Does the Relation Earn Its Label?

Week 11 Tuesday — Relations, equivalence, and partial orders

COMSC 2043 • Discrete Structures and Critical Thinking

October 27, 2026

Today's destination

- Say what a relation *is*: a set of ordered pairs.
- Test four properties one at a time, with a witness or a counterexample.
- Tell an **equivalence relation** from a **partial order**.
- Catch a confident AI claiming a property it does not have.

Warm-up: which of these is "the same as"?

1. "is friends with"
2. "has the same birthday as"
3. "is a sibling of"
4. "beats" in rock–paper–scissors

Gut check, hands up: for each one, is it a fair kind of "sameness"? Then name *one pair* that breaks it.

AI Fluency: a confident, untested label

An AI says

"Divisibility is *symmetric*, because 2 divides 6."

- What would symmetric actually require?
- Which single pair tests the claim?
- Write the missing property in one line.

What a relation is

A **relation** R on a set S is a set of ordered pairs $R \subseteq S \times S$.

- $a\,R\,b$ means $(a, b) \in R$. Order matters: $(2, 6) \neq (6, 2)$.
- Three ways to write the same thing: *list the pairs*, a *matrix*, a *digraph*.
- Name the domain S first. If the domain drifts, the answer changes.

Counting is humbling: on just 3 elements there are $2^9 = 512$ different relations.

The four properties, as questions

Reflexive	Does every a relate to itself? $(a, a) \in R$ for all a
Symmetric	Does every forward pair have a reverse? $(a, b) \Rightarrow (b, a)$
Antisymmetric	Do two-way pairs only happen for $a = b$? $(a, b) \wedge (b, a) \Rightarrow a = b$
Transitive	Do chains shortcut? $(a, b) \wedge (b, c) \Rightarrow (a, c)$

- To say **yes**: check *every* pair. To say **no**: show *one* witness.
- Symmetric and antisymmetric are not opposites. Both can hold; neither can hold.

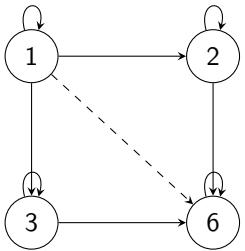
Model: divisibility on $S = \{1, 2, 3, 6\}$

$a \, R \, b$ when a divides b.

Pairs (9 of them):
$(1,1), (1,2), (1,3), (1,6),$
$(2,2), (2,6), (3,3), (3,6), (6,6)$

Matrix (row a, column b):

	1	2	3	6
1	1	1	1	1
2	0	1	0	1
3	0	0	1	1
6	0	0	0	1



Test each property separately

Property	Holds?	Witness / reason
Reflexive	yes	diagonal of matrix all 1s; $a \mid a$
Symmetric	**no**	$(2,6) \in R$ but $(6,2) \notin R$
Antisymmetric	yes	no off-diagonal pair is two-way (matrix has no mirrored 1s)
Transitive	yes	$(1,2),(2,6) \Rightarrow (1,6)$; all other chains checked

Verdict: reflexive + antisymmetric + transitive = a **partial order**. Not symmetric, so *not* an equivalence relation.

Your turn (14 min): Does the relation earn its label?

Use `student/tuesday-activity.md`.

1. Represent R as pairs **and** as a matrix or digraph.
2. Classify all four properties. Every "no" needs a counterexample pair or chain.
3. Equivalence, partial order, neither, or both? Name the required properties.
4. Check: does your matrix/digraph agree with your symmetry and antisymmetry answers?

Watch for: domain drift (using $\{1, 2, 3, 4, 5, 6\}$ by accident) and reversed pairs ($b \mid a$ instead of $a \mid b$).

Equivalence relation vs. partial order

	Equivalence	**Partial order**
Reflexive	✓	✓
Symmetric	✓	—
Antisymmetric	—	✓
Transitive	✓	✓
Feels like	*"is the same as"*	*"comes before / sits under"*

- Equivalence **sorts into bins** (a partition): "same remainder mod 3" makes 3 bins.
- Partial order **ranks, but not everything is comparable**: 2 and 3 do not divide each other.

Where you already meet these

Equivalence = "same bin"

- Same hash $\Rightarrow$ same file contents
- Same remainder mod n (clocks, checksums)
- Duplicate detection, grouping records

Partial order = "must come before"

- Prerequisites: which course before which
- Build order: `make`, package dependencies
- Git history: this commit descends from that one

Question for the room: a course scheduler treats prerequisites as a relation. Which property must it *never* violate to avoid a deadlock?

Challenge: how many bins?

Puzzle

Let $S = \{0, 1, \ldots, 11\}$ and $a \sim b$ when a and b have the same remainder mod 4.

1. How many bins? How big is each?
2. Is $\sim$ also a partial order? Give the property that fails.
3. Does $a \sim b$ and $b \sim c$ force $a \sim c$? Say why in one sentence.

Reasoning Odyssey Checkpoint 2 (Weeks 8–11)

- It **replaces** this week's Decision Gate. Submit one, not both.
- Pull from recursion, logic, probability, and relations: what changed in how you check a claim?
- Use the spine: **Sources** → **Rules** → **Work** → **Check** → **One-Sentence Summary**.

Plan it now (2 min): which one earlier claim of yours would you re-examine with a witness or counterexample?

Exit ticket (and Thursday)

Write one claim, one witness:

"Divisibility on $\{1, 2, 3, 6\}$ is _____ but not _____, because the pair _____ _____."

Thursday: Pair Reasoning. Bring a relation classifier, matrix, or digraph, and be ready to argue about it with a partner. The individual Pair Reasoning report is due after.